

Full-Stack Web Development — Day 9 Study Material

MODULE

React & UI Engineering

SUBJECT

Tailwind CSS, shadcn/ui & Sonner Toasts

AUTHOR

Penumuru Madhu Sudhan Reddy

Executive Summary: Comprehensive guide to utility-first styling with Tailwind CSS, copy-paste headless UI design using shadcn/ui, responsive data tables, dialog modal forms with inline field validation, and toast notification architecture with Sonner.

1. Tailwind CSS Utility-First Principles

Tailwind provides low-level utility classes directly in markup instead of writing custom CSS rules in separate stylesheet files.

Feature	Traditional CSS	Tailwind CSS
Location	Separated in .css files	Co-located in JSX className
Maintenance	Appends dead code over time	Unused classes automatically purged
Consistency	Arbitrary pixels (13px, 17px)	Design system scale (p-2, p-4)
Responsiveness	Manual @media queries	Mobile-first prefixes (sm:, md:, lg:)

2. shadcn/ui Architecture & Radix UI Primitives

Unlike component libraries installed as locked packages in node_modules, shadcn/ui is a collection of accessible components you copy directly into your project's repository.

- **Full Ownership:** You have full control over the component source code in components/ui/.
- **Accessibility First:** Built on accessible Radix primitives with WCAG AA compliance.
- **Conflict Resolution:** Uses cn() (combining clsx and tailwind-merge) to eliminate class conflicts.

3. Form Validation & Sonner Toast Notifications (Exercises 3 & 4)

```
// Form validation and Sonner notification trigger
const handleSubmit = (e: React.FormEvent) => {
  e.preventDefault();
  if (name.length < 3 || name.length > 100) {
    toast.error('Failed to save project. Name must be between 3 and 100 chars.');
```

Notification Feedback (o/p):

- On Valid Submit: "Project created successfully!" (Green toast alert)
- On Invalid Submit: "Failed to save project." (Red error toast alert)